

HANDS - ON RAG BOT & AGENTIC AI WORKSHOP

Build an EcoTourism RAG Chatbot + Agent Demo



LINK TO PRESENTATION

<https://github.com/boppanr/ScaleUpV5>

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Education:

- MIT Executive education in AI & Business Strategy(USA)
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WHAT IS RAG ?

RAG is crucial because LLM models, like ChatGPT, can sometimes "hallucinate" or provide outdated or incorrect information. By retrieving facts from trusted sources before generating responses, RAG ensures the answers are more reliable, up-to-date, and contextually relevant.

RAG DEMO

EcoTourism AI Assistant (Lightweight)

Ask about sustainable travel, eco-destinations, or green tourism tips!

 Chatbot

What is the top tourist season with the most visitors?

summer

Your question:

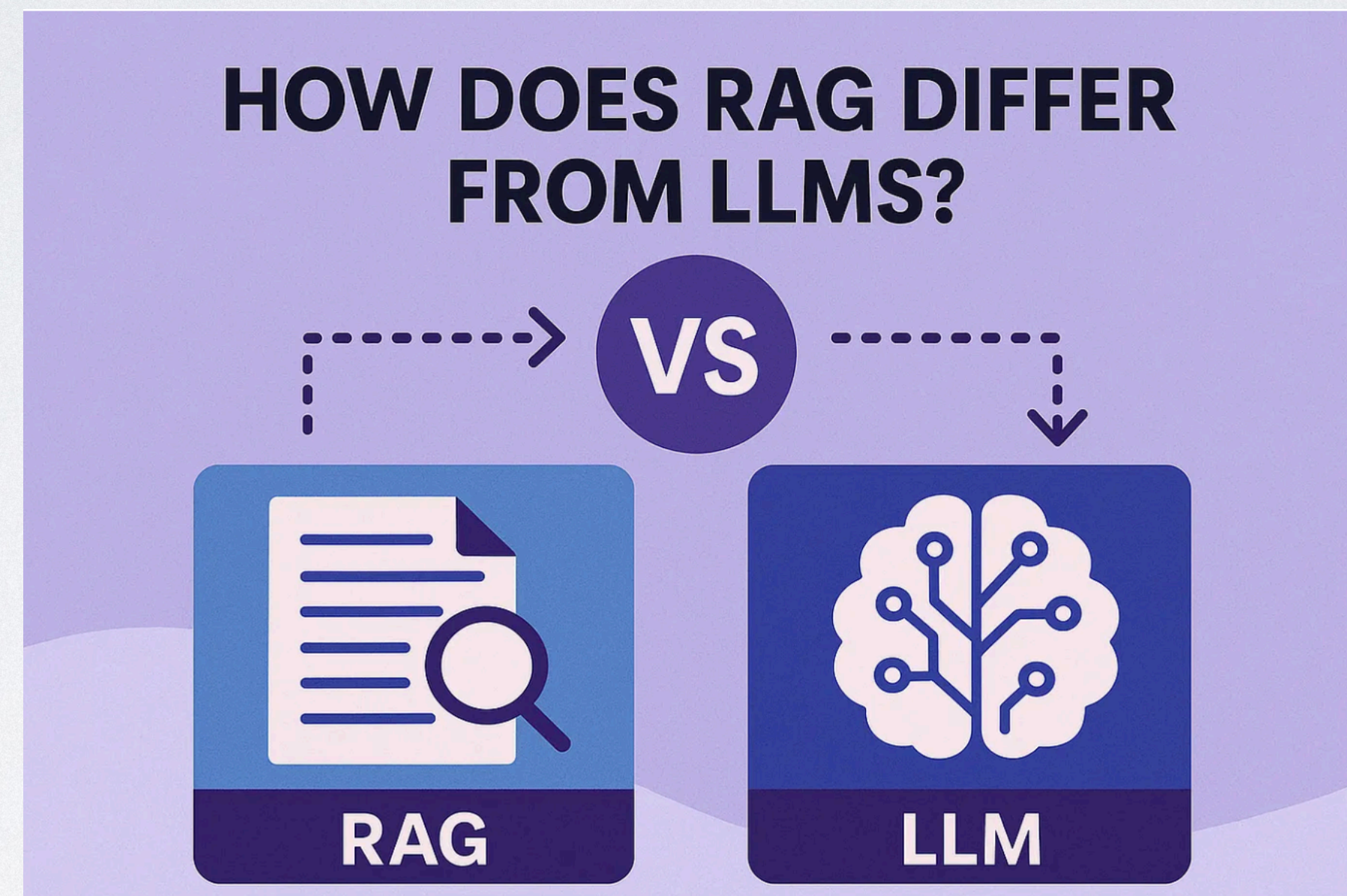
What is the top tourist season with the most visitors?

- What is the top tourist season with the most visitors?

WHAT WE ARE GOING TO DO TODAY ?

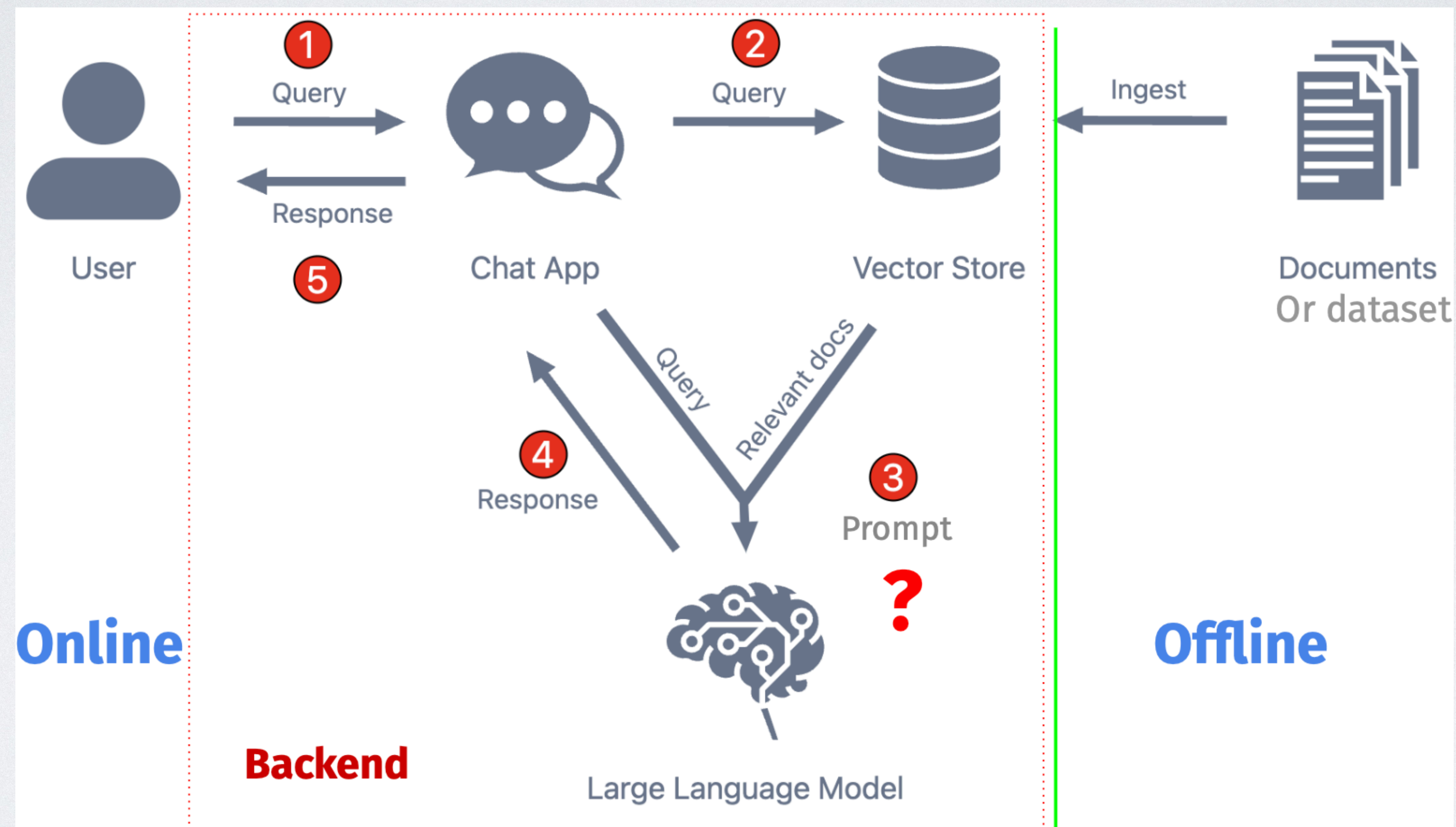
- Prepare Data
- Build Vector Store
- Wire RAG Pipeline
- Build Chatbot
- Create UI
- Test Queries

WHY RAG IS IMPORTANT



“RAG reduces hallucinations by grounding the LLM in real documents.”

RAG ARCHITECTURE





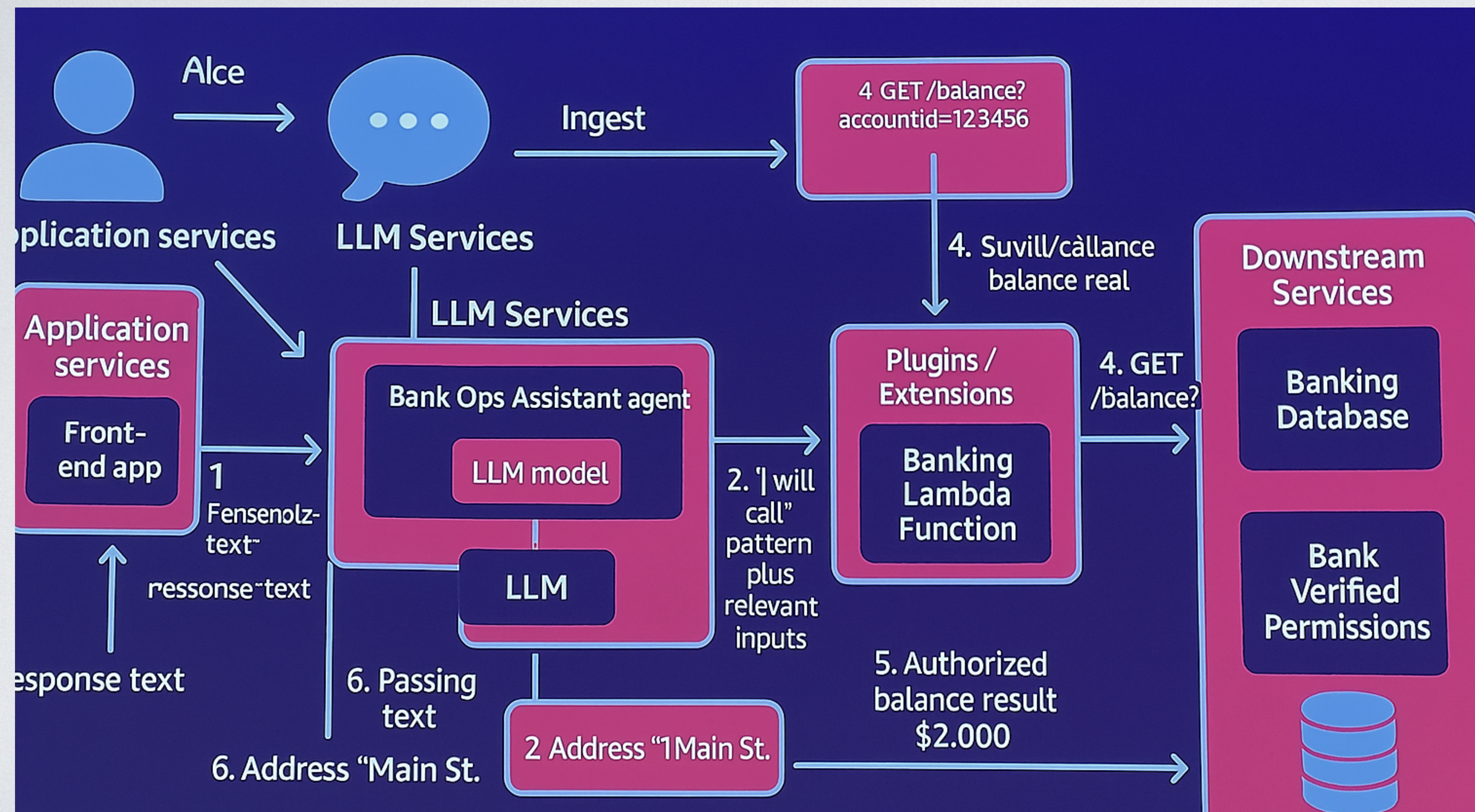
HANDS ON LAB(RAG BOT)

RAG EVALUATION

- Helpfulness
- Completeness
- Correctness
- Harmfulness (Responsible AI)



HOW TO SECURE A RAG / OWASP



Prompt Injection

Data Poisoning

Unauthorized Retrieval

Logging sensitive user data

QUESTIONS

HANDS - ON AGENTIC AI WORKSHOP

Build an Agent Demo

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AGENTIC AI DEMO

```
#  Sample user input request  
user_input = 'book 3PM slots for Dr. Lee'
```

```
#  Invoke the AI agent to process the request and execute the SQL operation  
output = agent_executor.invoke({"input": user_input})
```

I will now proceed with querying the database to book 3 PM slots for Dr. Lee.

```
[(23, 'Linda Lee', 'Dermatology', 8855162140)]
```

```
Invoking: `sql_db_query` with `{'query': "DELETE FROM slots WHERE doctor_id = 23 AND datetime = '2025-03-01 3:00 P  
M'"}`
```

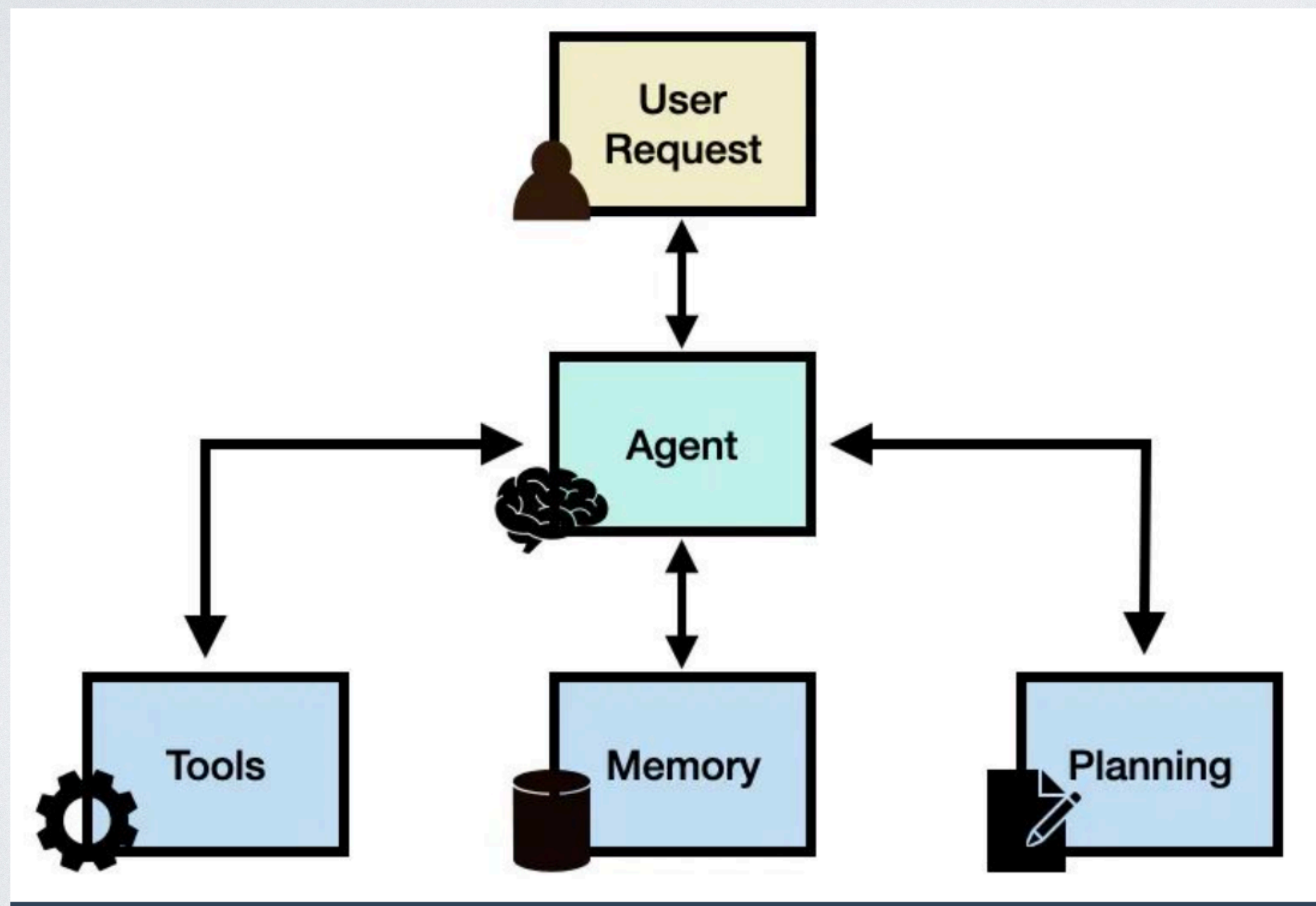
I have successfully booked the 3 PM slot for Dr. Linda Lee. The slot has been reserved.

> Finished chain.

WHY WE NEED AN AGENT

- Take Actions(Book an appointment, Book a flight etc)
- Make Decisions(Choose the best route)
- Interact with the real world.("Book a table at an Italian restaurant near me after 7 PM.")

AGENT ARCHITECTURE



Planner

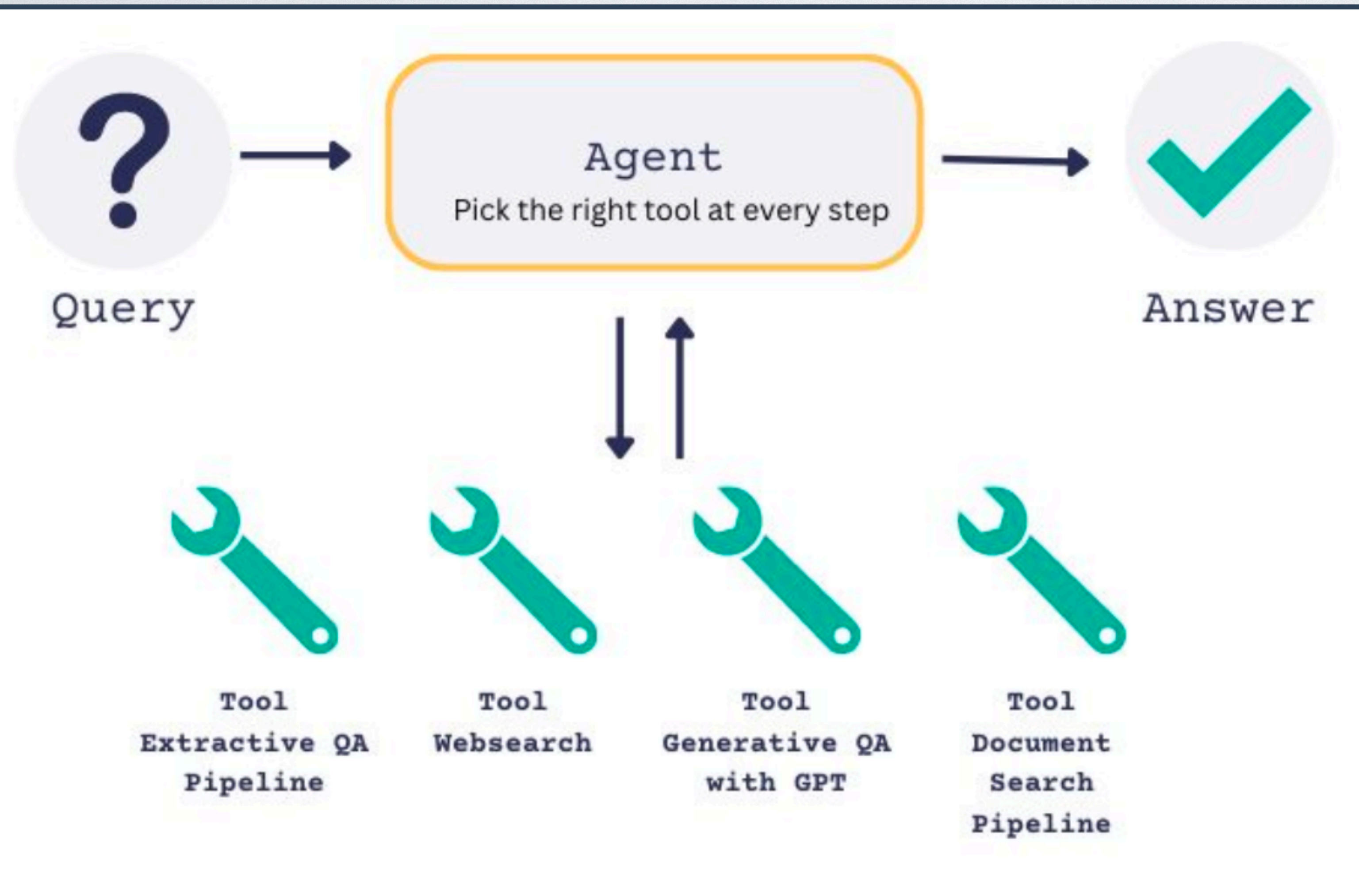
Tools

Memory

LLM

Executor

VARIOUS TYPES OF TOOLS THAT AGENT CAN CONNECT



Organize tools by category:

- Search
- Database
- API
- Utility functions



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HANDS ON LAB

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